

THE SCALAR ZETA SCATTERING SEMIGROUP IS NOT UNIFORMLY EXPONENTIALLY STABLE, UNCONDITIONALLY

OPENAI CODEX

ABSTRACT. Let ξ denote the completed Riemann zeta function, let $\Theta(s) = \xi(2s)/\xi(1+2s)$, which is inner on the right half-plane, and let $Z(t)$ be the truncated translation semigroup on the model space $K_\Theta = H^2 \ominus \Theta H^2$. We prove unconditionally that $\|Z(t)\| = 1$ for every $t \geq 0$, so the growth bound of Z is 0; equivalently, $\text{dist}_{H^\infty}(e^{-ts}, \Theta H^\infty) = 1$ for every $t > 0$. Under the Riemann hypothesis the point spectrum of the generator lies on the line $\text{Re } \lambda = -\frac{1}{4}$, so the spectral bound–growth bound gap of this semigroup is maximal *even assuming RH*. The mechanism is arithmetic: by Selberg’s theorem a positive proportion of zeta zeros lie on the critical line, and these zeros cluster with unbounded multiplicity in shrinking windows, producing explicit normalized reproducing kernels on which the semigroup is asymptotically norm-preserving. Consequently, no operator-norm form of the Lax–Phillips energy-decay criterion for RH can hold for any realization of the scattering semigroup that is norm-equivalent to this scalar model; the per-vector (local energy decay) form of the criterion is untouched.

1. INTRODUCTION

Lax and Phillips [3] and Faddeev–Pavlov [4] connected the Riemann hypothesis to the decay of the energy semigroup $Z(t)$ of the automorphic wave equation on the modular surface: RH is equivalent to exponential decay of $Z(t)f$ for initial data f of compact support, at rate governed by the real parts of the resonances, which sit at distance $\frac{1}{4}$ from the imaginary axis if and only if RH holds. It is tempting to formulate the criterion in operator norm,

$$(1) \quad \lim_{t \rightarrow \infty} \frac{1}{t} \log \|Z(t)\| \leq -\frac{1}{4} \quad (\text{purportedly equivalent to RH}),$$

and statements of this shape appear in the literature (e.g. Lax’s equation (54) in [5]).

The purpose of this note is to show that (1) fails *unconditionally* for the canonical scalar realization of the scattering semigroup: the truncated translation semigroup on the model space of the inner function $\Theta(s) = \xi(2s)/\xi(1+2s)$, whose zeros are the points $\rho/2$ over the nontrivial zeros ρ of ζ . Precisely, $\|Z(t)\| = 1$ for all t , so the uniform growth bound is 0 regardless of the truth of RH, while the spectral bound is $-\frac{1}{4}$ under RH. The proof is elementary once one observes that the on-line zeros guaranteed by Selberg’s theorem [1] cluster with unbounded multiplicity: clusters of m zeros make $|\Theta|$ decay like q^m at nearby points of small real part, and normalized reproducing kernels at such points are asymptotic eigenvectors of $Z(t)^*$ with eigenvalue modulus arbitrarily close to 1.

Since the argument uses only zeros *on* the critical line—which exist in both the RH-true and RH-false worlds—it cannot bear on RH itself. Its content is a correction of formulation: the RH-equivalent statement in the Lax–Phillips criterion is the per-vector decay of $Z(t)f$ for fixed admissible data, and any operator-norm strengthening of it is false. In semigroup-theoretic language, the scalar zeta scattering semigroup exhibits the maximal possible gap

between spectral bound and growth bound (cf. [9, 8]), with the gap forced by the arithmetic of the zero distribution. The phenomenon fits the general framework of spectral singularities of contractive semigroups of Naboko–Romanov [10], here with the singularity located at infinity.

Relation to the Lax–Phillips semigroup. Our theorem concerns the scalar model semigroup on K_Θ and is self-contained. Transferring the conclusion to the energy-space semigroup of [3] requires a unitary (or norm-) equivalence between that semigroup and the scalar compression; such an identification is developed by Uetake [6]. We do not re-derive it here, and we state the consequence for (1) conditionally on it (Corollary 5.3). Note that the full scattering matrix $S_c(s) = -\frac{s-1/2}{s+1/2} \Theta(s)$ differs from Θ by a single Blaschke factor, which changes neither the cluster structure of the zero set nor, therefore, the conclusion of the main theorem (Remark 5.4).

2. PRELIMINARIES

Let $\mathbb{C}_+ = \{s \in \mathbb{C} : \operatorname{Re} s > 0\}$ and let $H^2(\mathbb{C}_+)$, $H^\infty(\mathbb{C}_+)$ be the Hardy spaces of the half-plane. For $\mu \in \mathbb{C}_+$ the reproducing kernel of $H^2(\mathbb{C}_+)$ is

$$k_\mu(s) = \frac{1}{2\pi} \frac{1}{s + \bar{\mu}}, \quad \langle f, k_\mu \rangle = f(\mu), \quad \|k_\mu\|^2 = k_\mu(\mu) = \frac{1}{4\pi \operatorname{Re} \mu}.$$

For an inner function Θ on $\mathbb{C}_+$, the model space is $K_\Theta = H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+)$, with reproducing kernel

$$k_\mu^\Theta = P_{K_\Theta} k_\mu = k_\mu - \overline{\Theta(\mu)} \Theta k_\mu, \quad \|k_\mu^\Theta\|^2 = (1 - |\Theta(\mu)|^2) \|k_\mu\|^2.$$

For $\varphi \in H^\infty(\mathbb{C}_+)$ the truncated Toeplitz operator is $A_\varphi = P_{K_\Theta} M_\varphi|_{K_\Theta}$. We will use two standard facts (see [7, 2]):

- (i) $A_\varphi^* g = P_{H^2}(\bar{\varphi} g)$ for $g \in K_\Theta$, and K_Θ is invariant under the co-analytic Toeplitz operator $T_{\bar{\varphi}}$; moreover $T_{\bar{\varphi}} k_\mu = \overline{\varphi(\mu)} k_\mu$ for every $\mu \in \mathbb{C}_+$.
- (ii) (Sarason [2]) $\|A_\varphi\| = \operatorname{dist}_{H^\infty}(\varphi, \Theta H^\infty(\mathbb{C}_+))$.

The *truncated translation semigroup* on K_Θ is

$$Z(t) := A_{\varphi_t}, \quad \varphi_t(s) = e^{-ts} \quad (t \geq 0),$$

a strongly continuous contraction semigroup whose generator we denote A ; its point spectrum is $\{-\bar{a} : \Theta(a) = 0\}$, with eigenvectors of $Z(t)^*$ the kernels $k_a = k_a^\Theta$ at zeros a of Θ . (Adjoint and sign conventions differ across the literature; none affects operator norms, which is all we use.)

3. THE INNER FUNCTION

Lemma 3.1. *Let $\xi(w) = \frac{1}{2}w(w-1)\pi^{-w/2}\Gamma(w/2)\zeta(w)$ and*

$$\Theta(s) = \frac{\xi(2s)}{\xi(1+2s)} = \sqrt{\pi} \frac{2s-1}{2s+1} \frac{\Gamma(s)}{\Gamma(s+\frac{1}{2})} \frac{\zeta(2s)}{\zeta(2s+1)}.$$

Then Θ is inner on $\mathbb{C}_+$. Its zero set in $\mathbb{C}_+$ is exactly $\{\rho/2 : \zeta(\rho) = 0, 0 < \operatorname{Re} \rho < 1\}$, with multiplicities, and there is no cancellation: $\xi(1+\rho) \neq 0$ for every nontrivial zero ρ .

Proof. By the functional equation $\xi(w) = \xi(1-w)$ we have $\xi(1+2s) = \xi(-2s)$, so the zeros of the denominator in $\mathbb{C}_+$ would require $1+2s = \rho$, i.e. $\operatorname{Re} s = (\operatorname{Re} \rho - 1)/2 \in (-\frac{1}{2}, 0)$; hence Θ is analytic on the closed right half-plane (the trivial zeros of $\zeta(2s)$ lie at $\operatorname{Re} s < 0$ and are absorbed by Γ poles in ξ ; the apparent zero of $2s-1$ at $s = \frac{1}{2}$ cancels the pole of $\zeta(2s)$).

Zeros in $\mathbb{C}_+$: $\xi(2s) = 0$ iff $2s = \rho$, giving $s = \rho/2$ with $\operatorname{Re}(\rho/2) \in (0, \frac{1}{2})$; and $\xi(1 + \rho) \neq 0$ since $\operatorname{Re}(1 + \rho) > 1$.

Boundary modulus: since $\xi(\bar{w}) = \overline{\xi(w)}$ and $\xi(1 + 2i\tau) = \xi(-2i\tau) = \overline{\xi(2i\tau)}$, we get $|\Theta(i\tau)| = 1$ for all $\tau \in \mathbb{R}$ (ξ does not vanish on $\operatorname{Re} w = 0$).

Growth: for $\operatorname{Re} s \geq 2$ the factor $\zeta(2s)/\zeta(2s + 1)$ is bounded (absolute convergence) and $\Gamma(s)/\Gamma(s + \frac{1}{2}) = O(|s|^{-1/2})$ by Stirling, uniformly in $|\arg s| \leq \pi/2$; for $0 < \operatorname{Re} s < 2$ one has the finite-order bound $\zeta(2s) \ll (1 + |s|)^A$ and the classical lower bound $1/\zeta(\sigma + it) \ll (\log(|t| + 2))^7$ for $\sigma \geq 1$ [11, Ch. 3], so $|\Theta(s)| \ll (1 + |s|)^B$ on $\overline{\mathbb{C}_+}$. A function analytic on $\mathbb{C}_+$, continuous on the closure, of polynomial growth, with modulus ≤ 1 on the boundary is bounded by 1 on $\mathbb{C}_+$ by the Phragmén–Lindelöf principle for a half-plane. Hence $\Theta \in H^\infty(\mathbb{C}_+)$ with unimodular boundary values: Θ is inner. $\square$

4. CLUSTERS OF CRITICAL ZEROS

Lemma 4.1. *There exist $y_n \rightarrow \infty$, integers $m_n \rightarrow \infty$, and real numbers $\varepsilon_{n,1}, \dots, \varepsilon_{n,m_n}$ with $\max_j |\varepsilon_{n,j}| \rightarrow 0$, such that the points*

$$a_{n,j} = \frac{1}{4} + i(y_n + \varepsilon_{n,j}), \quad j = 1, \dots, m_n,$$

are distinct zeros of Θ .

Proof. By Selberg's theorem [1] the number $N_0(T)$ of zeros of ζ on the critical line with ordinate in $(0, T]$ satisfies $N_0(T) \geq cT \log T$ for some $c > 0$ and all large T . Cover $(0, T]$ by $\lceil T(\log T)^{1/2} \rceil$ windows of length $h_T = (\log T)^{-1/2}$. The average number of on-line ordinates per window is $\geq c'(\log T)^{1/2}$, so some window contains $m(T) \geq c'(\log T)^{1/2}$ distinct ordinates γ ; distinct on-line zeros have distinct ordinates, and each yields the zero $a = \frac{1}{4} + i\gamma/2$ of Θ by Lemma 3.1 (halving ordinates halves window lengths, which we absorb into h_T). Let y_T be the window center (after halving) and ε the offsets, so $|\varepsilon| \leq h_T/4 \rightarrow 0$. Finally $y_T \rightarrow \infty$: any bounded region contains finitely many zeros, while $m(T) \rightarrow \infty$. $\square$

Lemma 4.2. *Fix $\delta \in (0, \frac{1}{4})$ and let $\mu_n = \delta + iy_n$ with $y_n, m_n, \varepsilon_{n,j}$ as in Lemma 4.1. Then*

$$|\Theta(\mu_n)| \leq q_{n,\delta}^{m_n} \rightarrow 0 \quad (n \rightarrow \infty), \quad q_{n,\delta}^2 = \frac{(\frac{1}{4} - \delta)^2 + \max_j \varepsilon_{n,j}^2}{(\frac{1}{4} + \delta)^2 + \varepsilon_{n,j}^2} \xrightarrow{n \rightarrow \infty} \left(\frac{\frac{1}{4} - \delta}{\frac{1}{4} + \delta} \right)^2 < 1.$$

Proof. Let B_n be the finite Blaschke product over the cluster zeros $a_{n,j}$ (with $b_a(s) = \frac{s-a}{s+\bar{a}}$). Since these are zeros of the inner function Θ , we have $\Theta = B_n \Theta_n$ with Θ_n inner, so $|\Theta(\mu_n)| \leq |B_n(\mu_n)|$. For each factor,

$$|b_{a_{n,j}}(\mu_n)|^2 = \frac{(\frac{1}{4} - \delta)^2 + \varepsilon_{n,j}^2}{(\frac{1}{4} + \delta)^2 + \varepsilon_{n,j}^2} \leq q_{n,\delta}^2,$$

and there are m_n factors. Since $q_{n,\delta}$ is eventually bounded by some $q_\delta < 1$ and $m_n \rightarrow \infty$, the product tends to 0. $\square$

5. MAIN THEOREM

Theorem 5.1. *Unconditionally, $\|Z(t)\| = 1$ for every $t \geq 0$. Equivalently, the growth bound of the semigroup Z is $\omega_0 = \lim_{t \rightarrow \infty} t^{-1} \log \|Z(t)\| = 0$, and*

$$\operatorname{dist}_{H^\infty}(e^{-ts}, \Theta H^\infty(\mathbb{C}_+)) = 1 \quad \text{for every } t > 0.$$

Proof. Fix $t > 0$ and $\delta \in (0, \frac{1}{4})$, and let $\mu = \mu_n$ as in Lemma 4.2, writing $\theta = \Theta(\mu)$. Using $k_\mu^\Theta = k_\mu - \bar{\theta} \Theta k_\mu$ and the facts (i) of Section 2,

$$Z(t)^* k_\mu^\Theta = T_{\bar{\varphi}_t} k_\mu - \bar{\theta} T_{\bar{\varphi}_t}(\Theta k_\mu) = e^{-t\bar{\mu}} k_\mu - \bar{\theta} T_{\bar{\varphi}_t}(\Theta k_\mu),$$

where $\|T_{\bar{\varphi}_t}(\Theta k_\mu)\| \leq \|\varphi_t\|_\infty \|\Theta k_\mu\| = \|k_\mu\|$ because $\|\varphi_t\|_\infty = 1$ and multiplication by the inner function Θ is isometric on $H^2(\mathbb{C}_+)$. Hence

$$\frac{\|Z(t)^* k_\mu^\Theta\|}{\|k_\mu^\Theta\|} \geq \frac{e^{-t\delta} - |\theta|}{\sqrt{1 - |\theta|^2}} \quad (\text{using } \|k_\mu^\Theta\| = \sqrt{1 - |\theta|^2} \|k_\mu\|).$$

By Lemma 4.2, $|\theta| = |\Theta(\mu_n)| \rightarrow 0$ as $n \rightarrow \infty$, so $\|Z(t)\| = \|Z(t)^*\| \geq e^{-t\delta}$. Letting $\delta \downarrow 0$ gives $\|Z(t)\| \geq 1$; the reverse inequality holds since $Z(t)$ is a contraction. For a contraction semigroup, $\|Z(t_0)\| < 1$ for a single t_0 would force $\omega_0 \leq t_0^{-1} \log \|Z(t_0)\| < 0$ by submultiplicativity, so $\|Z(t)\| \equiv 1$ is equivalent to $\omega_0 = 0$. The distance statement follows from Sarason's formula (Section 2, (ii)). $\square$

Corollary 5.2. *Assume RH. Then every eigenvalue of the generator A has real part $-\frac{1}{4}$, while $\omega_0(Z) = 0$: the spectral bound–growth bound gap equals $\frac{1}{4}$, i.e. RH does not imply uniform exponential stability of the scalar model semigroup.*

Corollary 5.3. *Let $\tilde{Z}(t)$ be any C_0 -semigroup similar to $Z(t)$ with similarity constants $c\|f\| \leq \|Vf\| \leq C\|f\|$ (in particular, any unitarily equivalent realization, such as the identification of the Lax–Phillips energy semigroup with the scalar compression proposed in [6]). Then $\omega_0(\tilde{Z}) = 0$; in particular the operator-norm bound (1) fails for $\tilde{Z}$, unconditionally. This does not contradict the Lax–Phillips criterion in its per-vector form: for each zero a of Θ the eigenvector k_a satisfies $\|Z(t)^* k_a\| = e^{-t \operatorname{Re} a} \|k_a\|$, with $\operatorname{Re} a = \frac{1}{4}$ under RH, and $Z(t)f \rightarrow 0$ strongly for every $f \in K_\Theta$ since Θ is inner (the model semigroup is of class C_0).*

Remark 5.4. The full scattering matrix $S_c(s) = -\frac{s-1/2}{s+1/2} \Theta(s)$ is inner with the same zeros as Θ plus a simple zero at $s = \frac{1}{2}$. Lemmas 4.1–4.2 and the proof of Theorem 5.1 apply verbatim, so the conclusion holds for K_{S_c} as well.

Remark 5.5. The proof exhibits the mechanism concretely: normalized reproducing kernels $k_{\mu_n}^\Theta / \|k_{\mu_n}^\Theta\|$ at height y_n and abscissa δ are asymptotic eigenvectors of $Z(t)^*$ with asymptotic eigenvalue modulus $e^{-t\delta}$. Clusters of m_n zeros violate the Carleson separation condition with unbounded multiplicity, so the eigenvectors of $Z(t)^*$ do not form a Riesz basis and no uniform decay rate can be assembled from the eigenvalues. Equivalently, $|\Theta|$ fails to be bounded below on every strip $0 < \operatorname{Re} s < \delta$, so the resolvent of A is unbounded on every vertical line $\operatorname{Re} \lambda = -\delta$, and $\omega_0 = 0$ also follows from the Gearhart–Prüss theorem [8]. In the framework of [10], the semigroup has a spectral singularity at infinity.

Remark 5.6. Only the existence of $\gg T \log T$ zeros on the critical line is used, so Selberg's theorem suffices; neither simplicity (Levinson, Conrey) nor any information about off-line zeros enters. In particular the argument is identical whether RH holds or fails, which is why it can refute only the operator-norm *formulation* (1) and carries no information about RH itself.

REFERENCES

- [1] A. Selberg, *On the zeros of Riemann's zeta-function*, Skr. Norske Vid. Akad. Oslo I (1942), no. 10, 1–59.
- [2] D. Sarason, *Generalized interpolation in H^∞* , Trans. Amer. Math. Soc. **127** (1967), 179–203.

- [3] P. D. Lax and R. S. Phillips, *Scattering Theory for Automorphic Functions*, Annals of Mathematics Studies 87, Princeton Univ. Press, 1976.
- [4] L. D. Faddeev and B. S. Pavlov, *Scattering theory and automorphic functions*, J. Soviet Math. **3** (1975), 522–548.
- [5] P. D. Lax, [TODO: exact source of equation (54) — verify citation and whether the equation is stated in operator norm or per vector].
- [6] Y. Uetake, [TODO: exact title, journal, year 2007 — the unitary identification of the energy semigroup with the scalar compression].
- [7] S. R. Garcia, J. Mashregi, and W. T. Ross, *Introduction to Model Spaces and their Operators*, Cambridge Univ. Press, 2016.
- [8] J. Prüß, *On the spectrum of C_0 -semigroups*, Trans. Amer. Math. Soc. **284** (1984), 847–857.
- [9] K.-J. Engel and R. Nagel, *One-Parameter Semigroups for Linear Evolution Equations*, Springer GTM 194, 2000.
- [10] S. Naboko and R. Romanov, *Spectral singularities and asymptotics of contractive semigroups. I*, Acta Sci. Math. (Szeged) **70** (2004), 379–403. [TODO: verify volume/pages]
- [11] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford Univ. Press, 1986.